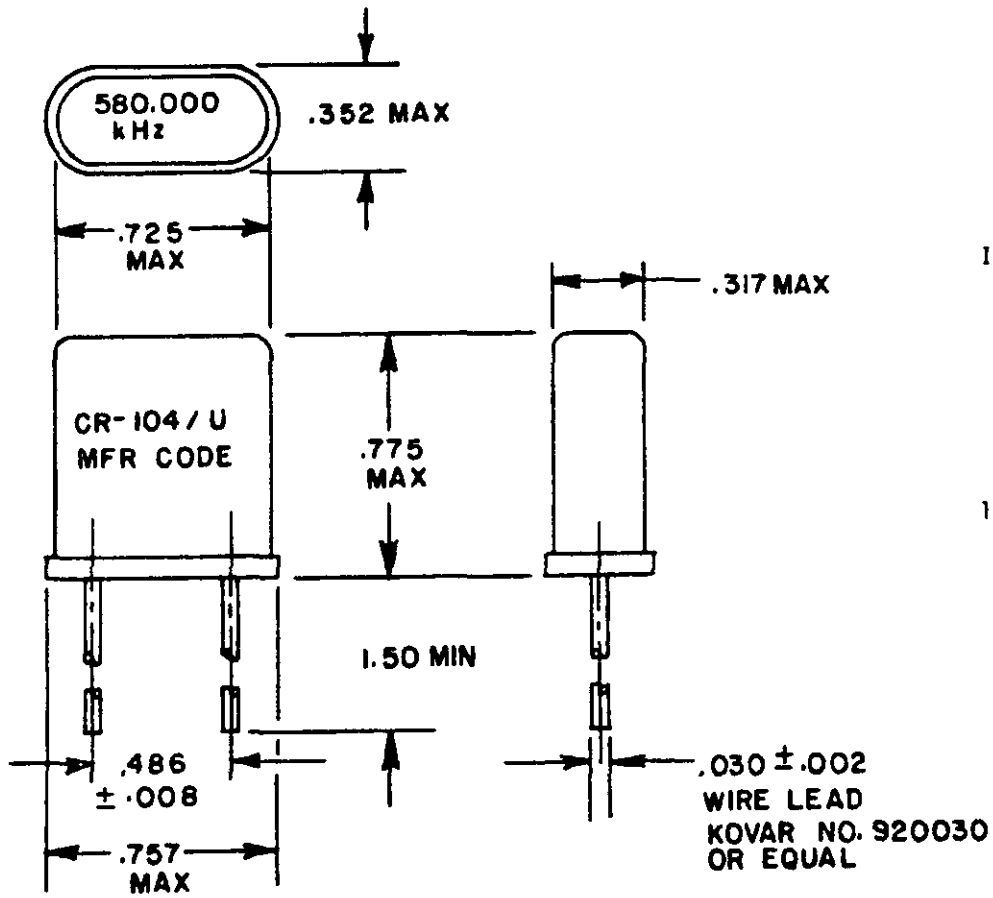
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-104/U

This specification sheet is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit, quartz, described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 200 to 580 kHz; fundamental; controlled; antiresonance.



INCHES	MM
.002	.05
.008	.20
.030	.76
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.68
1.50	38.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type designator, frequency and manufacturer's code shall not be less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. *Type CR-104/U, crystal unit.*

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-33/U (MIL-H-10056/20)
Dimensions, marking and configuration	See figure 1
Frequency range	200 to 580 kHz, inclusive
Mode of oscillation	Fundamental
Government reference standard test set	TS-710/TSM
Rated drive level	2.0 +0.4 mW
Calibration values	
Resistance	2000 Ohms
Resistor voltage drop	2.0 Volts
Antiresonance, load capacitance	21.5 ±0.5 pF
Reference temperature	75° ±1°C
Operating temperature (controlled)	70° to 80°C, inclusive
Frequency tolerance	See table II
Equivalent resistance	Frequency range, Resistance, Inclusive kHz: 200 to 300 — Maximum Ohms: 3,000; 300+ to 580 — 2,500
Frequency stability	±5 PPM
Operable temperature range	-40° to +70°C, inclusive
Shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±15 percent
Vibration: 1/	
Frequency change permitted	±5 PPm
Equivalent resistance change permitted	±15 percent
Thermal shock:	
Frequency change permitted	±10 PPM
Equivalent resistance change permitted	±15 percent
Bond strength:	
kHz	Grams, minimum
200 to 250	700
250+ to 320	500
320+ to 370	400
370+ to 435	300
435+ to 580	250

1/ Method 201 of MIL-STD-202, amplitude 0.015 inch (total excursion 0.030 inch).

TABLE II. Frequency response.

Frequency range, inclusive	Frequency tolerance
kHz	
200 to 339	±20 PPM
339+ to 484.999	±15 PPM
485 to 580	±10 PPM

1/ Crystal units in the frequency range 485 through 580 kHz (frequency tolerance ±10 PPM) shall be identified by by a blue lacquer dot approximately 1/8 inch in diameter. It shall be located on top of the can to the right of the letters "kHz."

Changes are not identified with respect to the previous issue due to the extensiveness of the changes.

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NOTE: These crystal units are for use in the RCA RT-594/ARC-38A equipment. They were previously fabricated using Air Force Technical Order 12-1-20 and/or RCA Drawing Number 8982273.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99

User activity:
Air Force - 17

Preparing activity:
Air Force - 85

(Project no. 5955-0606-3)

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MIL-C-3098/79C(USAF)

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